

AMAZON FRESH DATABASE ANALYSIS USING MYSQL

1. Problem Statement

An online grocery company wants to manage and analyze customer orders, products, suppliers, and reviews efficiently. The objective is to design a relational database using MySQL, establish relationships between entities, and perform SQL-based analysis to support business decision-making.

2. Dataset Collection

Dataset Files

- customers.csv
- products.csv
- suppliers.csv
- orders.csv
- order_details.csv
- reviews.csv

Columns in Dataset

Customerid,Productid,Orderid,Name,Age,Productname,Quantity,Unitprice,Total Price, Discount, Rating

Dataset Size

- Total Tables: 6
- Total Records:1000
- Total Attributes:42

3. Data Cleaning

Step 1: Checked Missing Values

- Identified NULL values
- Corrected missing records

Step 2: Removed Duplicates

- Used Excel filters and Remove Duplicates
- Verified uniqueness of primary keys

Step 3: Data Type Standardization

- CustomerID → VARCHAR(36)
- ProductID → VARCHAR
- OrderDate → DATE
- UnitPrice → DECIMAL

4. Database Design

Tables Created

- Customers
- Products
- Suppliers
- Orders
- Order_Details
- Reviews

Primary Keys

Table	Primary Key
Customers	CustomerID
Products	ProductID
Suppliers	SupplierID
Orders	OrderID
Reviews	ReviewID
Order_Details	(OrderID, ProductID)

Foreign Keys

Child Table	Foreign Key	Parent Table
Products	SupplierID	Suppliers
Order_Details	OrderID	Orders
Order_Details	ProductID	Products
Reviews	ProductID	Products

5. ER Diagram

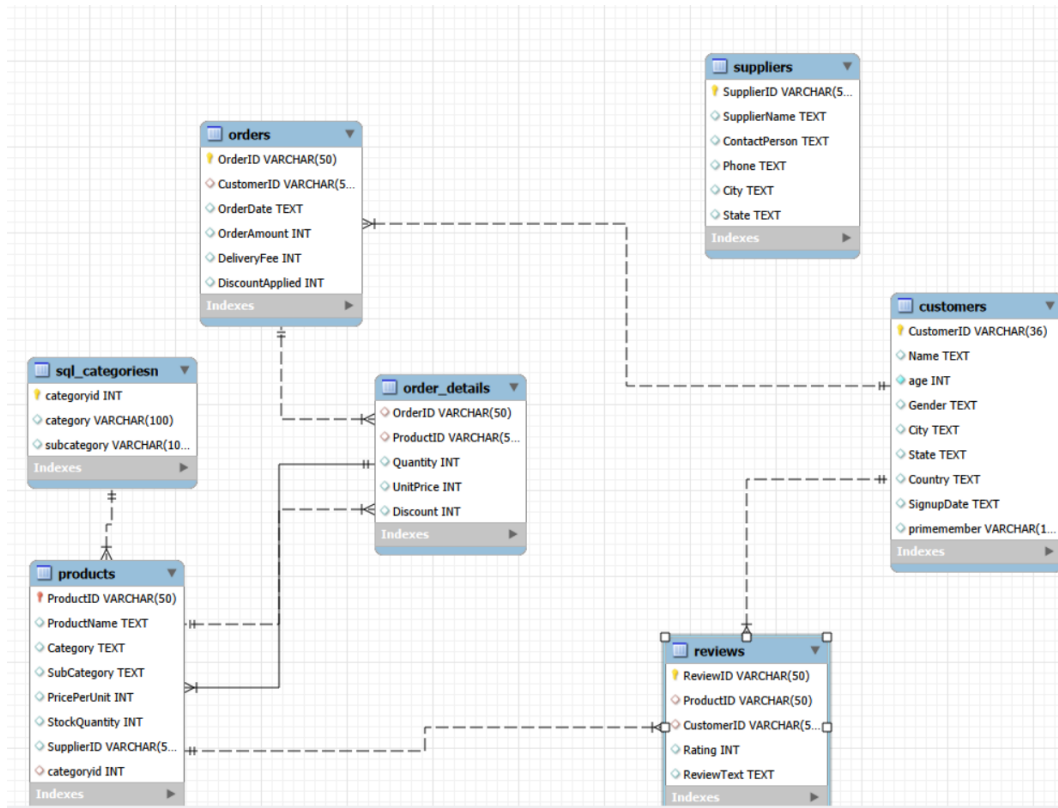


FIGURE 1:ER DIAGRAM

The ER diagram represents an Amazon Fresh database system designed to manage customers, products, suppliers, orders, reviews, and product categories. The Customers table stores customer information and is connected to the Orders table through CustomerID, establishing a one-to-many relationship where one customer can place multiple orders. The Orders table is linked to the Order_Details table, which acts as a junction table between orders and products, recording quantity, unit price, and discount for each product in an order. The Products table contains product details and is associated with both the Suppliers table and the Categories table, indicating that each product belongs to a supplier and a category. The Reviews table stores customer feedback and ratings, linking customers and products through CustomerID and ProductID. Overall, the database follows a normalized relational structure using primary and foreign key constraints to maintain data integrity, reduce redundancy, and support efficient analysis of Amazon Fresh business operations.

6. Data Analysis Using SQL Queries

Query 1: Fetch all products under the "Fruits" category.

```
select productname from amazon.products where category="fruits";
```

	productname
▶	Word Fruit
	Arrive Fruit
	Community Fruit
	Card Fruit
	Claim Fruit
	Fish Fruit
	Important Fruit
	Blue Fruit
	Four Fruit
	Wife Fruit
	Sell Fruit
	Report Fruit
	Popular Fruit

Figure 6.1: Category Of Fruits

Query 2:Join the Orders and OrderDetails tables to calculate total revenue per order.

```
select o.orderid,sum(c.Quantity*c.UnitPrice) as totalrevenue
```

```
from amazon.orders o
```

```
join amazon.order_details c
```

```
on o.OrderID=c.orderid
```

```
group by o.OrderID;
```

	orderid	totalrevenue
▶	000037f7-142c-4349-a419-51a985c4921b	14717
	00039c97-996c-4887-93f3-e42121c6452a	14667
	001432c4-ee5f-4bd5-8868-f83f035a804a	12069
	0015bb18-9c7d-4e85-9aef-c131b2e58615	3429
	0016ae7a-d034-48ae-970e-40be8f489297	6117
	003a7963-199e-4aa2-b1a0-426a596180e2	5140
	0048a700-055a-4107-b4c8-d4c64ad23192	332
	004cee1f-cf4b-4f90-a54d-68a3a2f85174	2211
	006f8f35-c056-4ef2-91dc-e224734c6cbc	8827
	00ad114c-166f-416a-a1fe-914b1b701abf	5848
	00bde197-eafa-4cb3-9bd5-346097128681	2016

Figure 6.2: Total Revenue

Query 3: HAVING clause to list products with average ratings greater than 4.

```
select productID,avg(rating) as avgrating from amazon.reviews
```

```
group by ProductID
```

```
having avg(rating)>4;
```

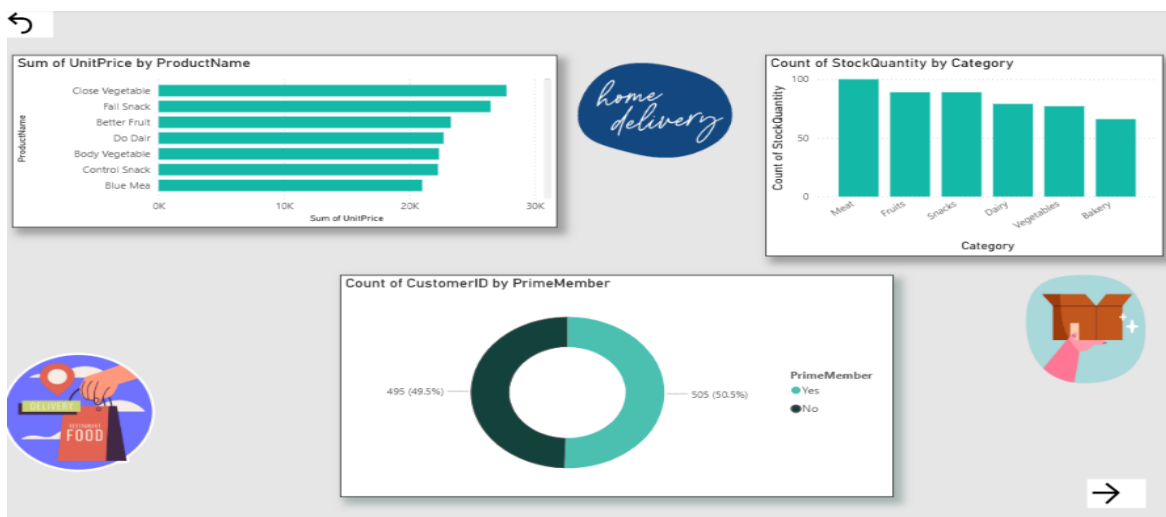
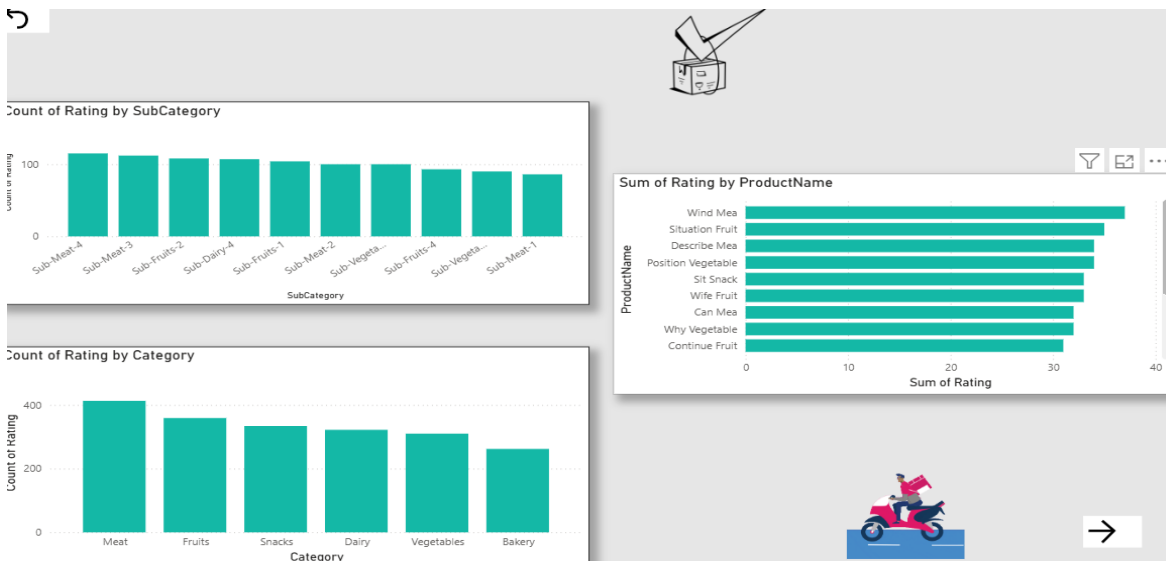
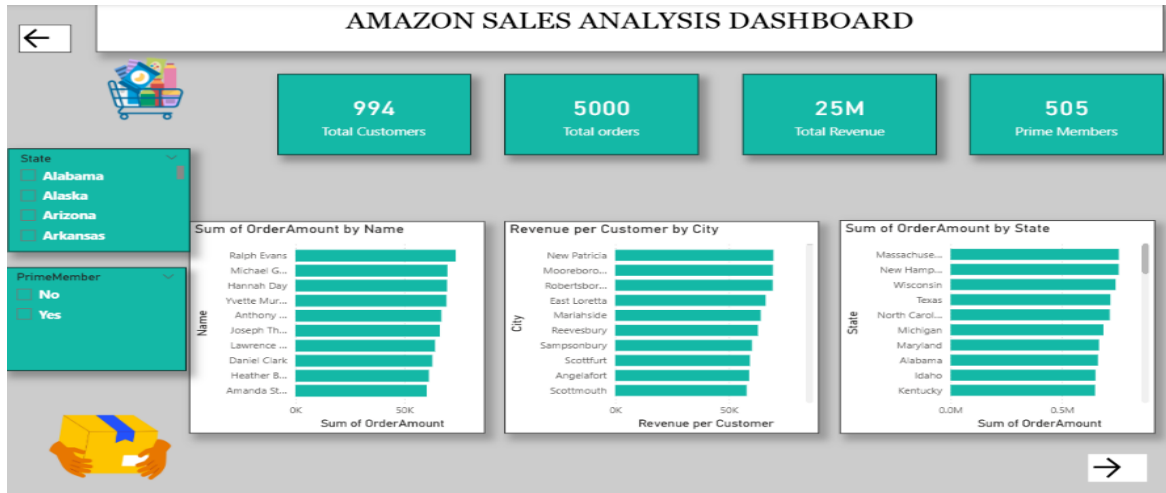
	productID	avgrating
▶	0ff68986-6dde-4494-bd75-88bac4160a7e	4.5000
	11dc08ec-ef6f-43d0-abaa-414a9c336956	4.2000
	145239de-d250-4d51-b078-eea3090c0db7	5.0000
	22fdf945-a880-4e03-aabc-35e691521ddd	4.3333
	2443e818-33af-4450-bab6-7cacc87e424	4.3333
	3233bde0-ced5-42e6-94c3-70502e4fd8c8	4.5000
	3812a751-a562-4541-a5b6-afa280950edd	4.3333
	46678d62-8c3f-4827-a27b-f7565f4c3e61	4.5714
	46c2e05c-1974-4783-90b9-a5b3885ce53e	4.3333
	54f7a232-2de5-46a4-a8df-d5e472835ba6	4.7500
	55a3d70a-c3c4-4a51-af63-ca3cc04eb799	5.0000
	63b5f8f4-6e55-42ad-98d4-bd6c3785d184	4.3333
	68-35588-4-8d-4-1f-6-4-788-f-1058b-2	4.1428

Figure 6.3: Average Rating

7. Database Implementation

- Imported CSV files using Table Data Import Wizard.
- Assigned appropriate data types.
- Added Primary Keys and Foreign Keys.
- Established table relationships.

8. Dashboard Development and Visualization



9. Conclusion

The Amazon Fresh database was successfully designed and implemented using MySQL Workbench. The dataset was cleaned, imported, and organized into multiple related tables including Customers, Orders, Products, Suppliers, Reviews, and Order Details. Appropriate primary key and foreign key constraints were applied to establish relationships and maintain data integrity. An ER diagram was created to visualize the database structure and relationships between entities. Various SQL queries were performed to analyze customer behavior, product performance, supplier information, and order details. In addition, an interactive dashboard was developed to present key insights and business metrics in a clear and visual manner, enabling effective data analysis and supporting informed decision-making. Overall, the project demonstrates the practical application of database design, data management, SQL querying, and data visualization techniques in an e-commerce environment.